

Paragliding Intelligence Platform

Volume 11 - Business Plan v2 Technical

Commercial strategy, revenue architecture, SaaS model, go-to-market, financial assumptions, KPI system, and investor milestones

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Relationship to previous volumes	Depends on Volumes 01-10: PRD, SAD, Database, API, Weather Engine, AI/ML, Mobile UX, Infrastructure, Security, QA
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Important safety positioning:

This business plan assumes the product is a decision-support and briefing platform. It must not be marketed as a flight authorization system. Pilots remain responsible for final decisions, and all safety-related paid features must preserve uncertainty, confidence and NO DATA states.

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1. Executive Business Thesis

The Paragliding Intelligence Platform is not positioned as another weather map. The core business thesis is that pilots, schools, clubs and event organizers do not primarily pay for prettier weather layers; they pay for reduced uncertainty, better timing, safer decisions, less wasted travel, and operational coordination. The platform converts weather data, terrain data, live observations, pilot profiles and historical launch behavior into paragliding-specific decisions: GO, MAYBE, NO-GO, best launch window, confidence, risk drivers and recommended alternatives.

The commercial wedge is a vertical decision-intelligence product for free-flight. Windy and Ventusky are broad weather visualization platforms. XCTrack is a strong flight-computer/navigation tool. Paragliding Map is strong for site discovery and site metadata. The opportunity is to combine these categories into a paragliding-first operational platform that owns the pre-flight decision, in-flight safety alert, club/school workflow and post-flight learning loop.

Strategic business goal

- Create the default pre-flight and launch-site decision app for paragliding pilots.
- Use free weather/site discovery to acquire users, then convert high-intent pilots to paid decision tools.
- Build defensibility through local digital twins, pilot reports, forecast error history, launch playbooks and weather-station data.
- Expand from individual subscriptions into clubs, schools, competition events, rescue tooling and API licensing.
- Maintain safety trust by never hiding uncertainty behind a false-green recommendation.

Why this can become stronger than general weather apps

A general weather app optimizes for visualizing weather variables. This platform optimizes for a pilot action: whether to drive, where to launch, when to launch, whether to wait, when to land and how conditions compare with the forecast. The decision layer is the monetizable product.

Differentiator	Business value	Defensibility
Launch-specific forecast correction	Improves trust and willingness to pay	Requires historical forecast errors, station data and reports per launch
Flyability and safety scores	Turns complex weather into a decision	Requires paragliding domain rules and QA
Drive-or-dont-drive engine	Saves time and fuel for pilots	Requires user location, drive window and site forecast ranking
Club/school dashboards	Creates B2B recurring revenue	Operational workflows and member/students data create switching costs
Live reality layer	Makes app useful on the day of flight	Crowd data and weather stations improve with scale
Digital twin launch models	Improves local accuracy over time	Data moat compounds per site and region

Revenue summary

The model combines B2C subscriptions, B2B club/school subscriptions, event packages, rescue/authority contracts, API access, marketplace leads and hardware weather-station revenue. The recommended near-term focus is B2C Pro/Pro+ plus club and school accounts. The long-term platform value comes from combining recurring subscriptions with proprietary local weather intelligence.

2. Product-Market Problem

Paragliding pilots face a decision problem, not merely a weather-access problem. They need to combine wind direction, wind speed, gust spread, cloudbase, thermals, rain, storms, valley wind, terrain, launch orientation, pilot skill, traffic time and local site knowledge. Existing tools usually require pilots to mentally assemble data across several apps and local chat groups. This creates friction, wasted travel, inconsistent decisions and preventable risk.

Core jobs to be done

User job	Current workaround	Business opportunity
Know if a site is worth driving to	Check Windy, local station, webcams, chat groups	Sell a ranked drive-or-dont-drive recommendation
Know if a launch is safe now	Use generic wind/gust data and local experience	Sell live launch-specific risk score and confidence
Find a flyable site nearby	Manual map browsing and local knowledge	Sell ranked site discovery by pilot level and time window
Plan XC route	Use XCTrack/SeeYou/etc plus weather maps	Sell XC route weather optimization and landing-field intelligence
Run school/tandem operations	Instructor judgment, scattered forecasts, chat	Sell daily briefing and student risk dashboard
Run club safety and events	Manual coordination and weather officer	Sell club dashboard and event control center

Pain severity

The pain is highest on marginal days, mountain sites and sites with strong local microclimate. These are exactly the days where pilots need a trustworthy decision layer. The paid product should therefore not focus only on fair-weather visualization; it should own uncertainty, disagreement and risk. A forecast confidence engine should be a premium value driver.

Safety and trust requirement

The product should never promise certainty. The safest commercial language is: decision support, risk briefing, forecast confidence and site intelligence. Avoid language such as “safe to fly” without context. Use GO/MAYBE/NO-GO only with confidence, uncertainty and hard-blocker explanations.

3. Market Context and Evidence Base

This business plan uses a conservative market model because exact global active-pilot counts are not consistently published by a single authoritative source. Equipment-market reports and official competition/association references indicate a meaningful and growing free-flight ecosystem, but the startup should validate the serviceable market through early pilots, clubs, schools and regional launches rather than relying on inflated top-down market estimates.

Current evidence

Evidence	Interpretation for business plan
Paragliding equipment market reports estimate hundreds of millions USD in annual global equipment sales and mid-single-digit to high-single-digit CAGR.	The ecosystem is large enough for a specialized subscription SaaS, but market reports vary widely; use them as directional evidence only.
FAI/CIVL continues to operate international hang gliding and paragliding competitions.	There is an organized competition segment with high need for weather, safety and tracking workflows.
Paragliding Map advertises global site, landing, station and webcam coverage.	Site discovery and live site intelligence is already validated demand; differentiation must be decision intelligence and local correction.
XCTrack positions itself as an advanced Android flight computer for PG/HG with competition, airspace, FAI assistant and tracking features.	In-flight navigation is a strong incumbent category; the new platform should integrate, not duplicate blindly, and win pre-flight and operations workflows.
Windy and Ventusky offer broad weather layers, models, radar/satellite and premium plans.	Users already pay for weather visualization; this product must justify a higher vertical value through paragliding-specific decisions.

Market segmentation method

Instead of starting with a single TAM number, the plan uses a bottom-up segment model: individual active pilots, serious XC pilots, schools/tandem operators, clubs, competitions/events, rescue/authorities, API customers and hardware/weather-station customers. Each segment has its own willingness to pay, acquisition motion and support requirements.

Segment	Estimated willingness to pay	Sales motion	Primary product
Weekend pilot	Low to moderate	Self-serve app store/web	Pro subscription
Serious XC/competition pilot	Moderate to high	Self-serve + ambassador	Pro+ subscription
Club	Moderate recurring	Founder-led sales then self-serve	Club dashboard
School/tandem operator	High recurring	Direct sales	School operations dashboard
Competition/event	High one-off/seasonal	Direct sales	Event weather center
Rescue/authority	Enterprise	Relationship sales	Rescue mode / incident tools
API/hardware partners	Variable	Partner sales	Forecast/flyability APIs and station dashboards

Source notes

The document sources public product positioning from Windy, Ventusky, XCTrack and Paragliding Map. It also uses market research estimates as directional evidence only, because such reports often differ materially in methodology and numbers.

4. Competitor and Positioning Analysis

Competitor map

Competitor/category	Strength	Weakness/opportunity for this platform
Windy	Excellent broad weather visualization, many layers, premium forecast/radar features	Not paragliding-specific decision intelligence; users must interpret data themselves
Ventusky	Broad model/layer approach, radar/satellite/webcam positioning	General weather app; not built around launch orientation, pilot skill or site-specific flyability
XCTrack	Strong Android flight computer, tracking, airspace, competition support	Less focused on pre-flight weather decision, paid SaaS club/school operations and forecast correction
Paragliding Map	Strong site database, weather stations, webcams and site discovery	Opportunity to add paid GO/MAYBE/NO-GO intelligence, digital twins and operational workflows
Local WhatsApp/Telegram groups	High trust and local nuance	Fragmented, not structured, not forecast-backed, hard to monetize but important for community acquisition

Positioning statement

For paragliding pilots, schools, clubs and events who need reliable launch decisions, the Paragliding Intelligence Platform is a vertical weather-decision system that combines forecasts, terrain, live reality, site rules, pilot skill and AI briefings into actionable risk-aware recommendations. Unlike general weather maps, it learns site-specific errors and explains why conditions are GO, MAYBE or NO-GO.

Category strategy

Category choice	Why it matters
Avoid being "another weather app"	Weather visualization is crowded and users already have defaults.
Create "paragliding decision intelligence" category	A new category allows higher willingness to pay than generic weather.
Own the pre-flight decision moment	The user checks the app before driving; this is high-intent and monetizable.
Become operational software for schools/clubs/events	B2B subscriptions improve revenue predictability.

Pricing reference

Public competitor subscription prices are not always consistently exposed by region or platform. This plan uses competitor features and approximate public premium-plan positions as context, but pricing should be validated with willingness-to-pay tests in the pilot community. Price anchoring should be based on saved travel costs, safety confidence and school/club operational value, not merely weather-layer access.

5. Customer Segments and Personas

Persona	Primary need	Risk/constraint	Commercial path
Beginner pilot	Needs simple safe/no-go guidance and learning explanations	High false-green sensitivity, low tolerance for technical weather	Free -> Pro via safety and learning
Weekend recreational pilot	Wants to know whether a site is worth the drive	Hates wasted days and scattered data	Pro annual plan
XC pilot	Needs thermals, cloudbase, model confidence, route options	Pays for better planning and post-flight learning	Pro+ annual plan
Competition pilot	Needs task weather, airspace, route ranking, in-flight alerts	Requires reliability and low latency	Pro+ / Event
Club officer	Needs member safety, site status, verified reports, announcements	Needs moderation and admin tools	Club subscription
School/tandem operator	Needs daily operational briefing and student suitability	High liability and schedule pressure	School subscription
Event organizer	Needs task/weather center, tracking, safety monitor	High stakes for a limited period	Event package
Rescue/authority	Needs last-known-position, wind drift and search-area exports	Requires privacy and verified access	Enterprise/rescue contract

Persona-driven feature priorities

MVP should prioritize weekend pilots and serious local pilots because they create daily app usage and community data. The first B2B motion should target clubs and schools in launch-dense regions, because they already coordinate weather decisions and can act as distribution hubs.

Persona	MVP must-have	Premium trigger
Weekend pilot	Favorite sites, forecast, flyability score, basic alerts	Unlimited alerts, drive recommendation, confidence, 10-day view
XC pilot	Thermal/cloudbase layers, XC score	Route planner, model comparison, AI XC briefing, post-flight debrief
Club	Site status and member reports	Club dashboard, verified reports, station management
School	Daily site conditions	Instructor dashboard, student skill matching, exportable briefing

6. Monetization Architecture

The monetization architecture must be implemented as entitlements at the platform level, not as UI-only locks. Every backend response should know the user plan, account context, region, quota and feature flags. This prevents accidental access leakage and enables fast experimentation.

Revenue streams

Stream	Description	Priority	Notes
B2C subscriptions	Free, Pro, Pro+ for individual pilots	High	Primary early revenue and validation path
Club subscriptions	Shared dashboard, member reports, site management	High	Good early B2B wedge through local communities
School/tandem subscriptions	Instructor dashboard, student/flight suitability, daily reports	High	Higher ACV and operational pain
Event packages	Weather center, tracking, task planning, safety monitor	Medium	Seasonal but high visibility
API access	Forecast/flyability/site APIs for partners	Medium	Requires mature data quality and support
Rescue/authority contracts	Search-area, drift and incident tools	Low early, high strategic	Requires trust, compliance and relationship sales
Marketplace leads	Schools, guides, tandem flights, shops	Medium after user base	Can generate non-subscription revenue
Weather-station hardware	Station sales and station dashboard subscriptions	Medium	Strong data moat but ops complexity

Entitlement architecture

Entitlement object returned by /me/entitlements:

```
{
  "plan": "PRO_PLUS",
  "account_type": "individual",
  "features": {
    "forecast_days": 10,
    "model_comparison": true,
    "ai_briefings_per_day": 20,
    "xc_planner": true,
    "offline_region_packs": 5,
    "live_tracking_minutes_per_month": 3000,
    "alert_rules": 50
  },
  "billing_provider": "stripe|apple|google|manual_invoice",
  "expires_at": "2027-06-04T00:00:00Z"
}
```

Key monetization principle

The free product should be genuinely useful, but it should stop before advanced decision automation. Free users can see weather and site basics. Paid users get confidence, ranked recommendations, advanced alerts, model comparison, XC tools, AI briefings, offline packs and personal risk calibration.

7. Packaging, Pricing and Entitlements

Pricing below is a starting hypothesis. It should be validated with interviews, landing pages and in-app paywall experiments. Use annual plans as the default because paragliding has seasonal usage and pilots are more likely to pay once per flying season than monthly every winter.

Tier	Target	Suggested price	Core features	Conversion trigger
Free	All pilots	EUR 0	3-day site forecast, basic map, 3 favorites, community reports, limited alerts	Discovery and trust building
Pro	Weekend pilots	EUR 5.99/mo or EUR 59/yr	10-day forecast, hourly flyability, unlimited favorites, advanced alerts, drive-or-dont-drive, forecast confidence	Save one wasted trip
Pro+	XC and serious pilots	EUR 11.99/mo or EUR 119/yr	Model comparison, XC planner, AI briefing, advanced thermal/cloudbase, offline packs, post-flight debrief	Plan better XC days
Club	Clubs	EUR 39-99/mo	Club dashboard, member reports, site status, announcements, station dashboard	Coordinate site safety
School	Schools/tandem	EUR 129-299/mo	Instructor dashboard, student suitability, daily briefing, exportable reports	Operational reliability
Event	Competition/festival	EUR 500-3000/event	Weather center, task decision support, tracking, safety board	Run safer events
Rescue	Authorities	Custom	Incident tools, drift, search area, data export	Emergency response value
API	Partners	EUR 99-999/mo + usage	Forecast, site, flyability and risk APIs	Integrate decision intelligence

Feature entitlement matrix

Feature	Free	Pro	Pro+	Club/School/Event
Favorite sites	3	Unlimited	Unlimited	Organization-level
Forecast horizon	3 days	10 days	10+ days where licensed	Operational dashboard
Flyability score	Basic	Full with hourly details	Full + XC context	Group/site dashboard
Model comparison	No	Limited	Yes	Yes
AI briefing	3/week	5/day	20/day	Org briefing templates
XC planner	No	Limited preview	Full	Event task planning
Offline maps	No	1 region	5 regions	Org packs
Live tracking	Limited	Personal	Extended	Team/event/rescue views
Weather station admin	No	No	No	Yes

Paywall rules

- Never hide safety blockers behind a paywall. If there is a hard NO-GO risk, all users should see it.
- Do not use fear-based paywalls. Monetize convenience, confidence, advanced planning and operations.
- Always show why a feature is locked and the value it would add for the current site/day.
- Allow seasonal pause or annual renewal reminders to reduce churn in winter regions.

8. Go-to-Market Strategy

Beachhead strategy

Start in a launch-dense region where users already check multiple tools and local conditions vary by site. Norway is a useful personal starting market for field validation, but the strongest early growth regions are likely the Alps and other high-density European paragliding areas because they contain many active clubs, schools, webcams, stations and cross-country pilots.

Phase	Region focus	Goal	Success metric
Alpha	1-3 local Norwegian sites	Validate scoring logic and UI with real pilots	20 pilots use weekly; no false-green incidents in review
Private beta	Norway + one Alpine valley	Validate site ranking and reports	500 registered users; 100 weekly active users
Public MVP	Nordics + Alps selected sites	Validate Pro conversion	5,000 users; 3-5% paid conversion
V1 commercial	Europe launch regions	Scale subscriptions and club/school sales	25,000 users; EUR 20k+ MRR
V2 global	Europe + selected global hot spots	Expand site database and language support	100,000 users; B2B revenue >25% of MRR

Launch motion

26. Recruit 20 trusted local pilots as founding testers.
27. Build 10 golden launch playbooks with local rules, wind limits, webcams and known hazards.
28. Publish daily flyability briefings for pilot communities during the season.
29. Use transparent forecast-vs-reality posts to build trust.
30. Offer free club dashboards to the first 5 clubs in exchange for feedback and testimonials.
31. Create regional landing pages optimized for “paragliding weather [site]” searches.

Go-to-market channels

Channel	Tactic	Metric
Local clubs	Free trial club dashboards and weather officer tools	Club activation rate
Schools	Instructor dashboard pilots and tandem briefing exports	School conversion rate
YouTube/short video	Daily condition breakdowns and forecast-vs-reality	Install conversion
SEO	Site forecast pages indexed by location	Organic signups
Events	Sponsor weather center for selected competitions	Pro+ conversions
Ambassadors	Trusted XC pilots publish route briefings	Referral rate
Weather stations	Club station program	Station data coverage

9. Growth Loops and Distribution

Data growth loop

The most important growth loop is data-driven: more users produce more observations, which improves launch reality and forecast correction, which creates better recommendations, which attracts more users. This loop must be designed into the product from the start.

User checks site -> user submits report -> report validates forecast ->

digital twin improves local correction -> next users get better recommendation ->

trust increases -> more users report and subscribe.

Viral and network loops

Loop	Product mechanic	Business value
Shareable briefing	Share a site briefing link to club chat	Acquires pilots organically
Live tracking link	Share flight with friends/rescue contacts	Creates non-pilot awareness and safety trust
Club report	Verified local pilot report appears on site page	Creates community status and habit
Event weather board	Public event forecast dashboard	High-visibility marketing
Post-flight debrief	Pilot shares AI debrief screenshot	Promotes Pro+

SEO moat

Each launch site can become a search landing page containing site metadata, wind directions, weather summary, webcam preview, flyability status and local playbook. The public page should provide enough value to rank and convert, while advanced forecasts and decision tools require signup.

10. Partnerships and Ecosystem Strategy

The platform should not try to replace every existing tool. The strongest path is to integrate with flight computers, site databases, clubs, schools, event systems, weather-station providers and airspace datasets.

Partner type	Value to platform	Value to partner	Integration
Paragliding clubs	Local trust, reports, site rules	Better safety and coordination	Club dashboard, verified officer role
Schools/tandem operators	Paid B2B and daily usage	Briefing and scheduling efficiency	Instructor dashboard, export reports
Weather stations	Live reality layer	Dashboard and maintenance tools	MQTT/HTTP station ingest
Flight apps/devices	Tracking and IGC data	Better weather intelligence	API/export integrations
Competitions	Visibility and high-value use case	Weather center and safety board	Event API and tracking
Airspace data providers	Safety-critical data	Traffic and attribution	Airspace import pipeline
Equipment shops	Marketplace revenue	Qualified local leads	Marketplace listings

Partner sales principles

- Offer clubs data ownership transparency and export options.
- Do not compete with schools; help schools acquire and retain students.
- Make station integrations vendor-neutral to avoid hardware lock-in too early.
- Keep safety-related data attribution clear and auditable.

11. Financial Model Assumptions

The financial model is intentionally conservative and staged. It uses bottom-up assumptions for users, paid conversion, annual subscription price, B2B accounts and operating cost. Actual performance must be measured with a live cohort model after MVP launch.

Base-case user assumptions

Metric	Year 1	Year 2	Year 3	Year 5
Registered users	5,000	25,000	75,000	250,000
Monthly active users	1,500	9,000	30,000	110,000
B2C paid conversion	3%	5%	7%	9%
Paid B2C users	150	1,250	5,250	22,500
B2C blended ARPU/year	EUR 70	EUR 82	EUR 92	EUR 100
Club accounts	3	25	80	250
School accounts	1	10	35	120
Event packages	0	5	20	80
API customers	0	2	8	30

Core pricing assumptions

Product	Annualized base assumption	Comment
Pro individual	EUR 59/year	Main weekend-pilot product
Pro+ individual	EUR 119/year	Serious/XC pilots
B2C blended ARPU	EUR 70-100/year	Mix shifts toward Pro+ over time
Club	EUR 70/month average in early years	Mix of small and larger clubs
School	EUR 180/month average in early years	Tandem/school operational value
Event	EUR 1,200/event average	Weather center and tracking bundle
API	EUR 250/month average early	Small partner integrations

Sensitivity assumptions

Variable	Low case	Base case	High case	Management lever
Paid conversion	2-4%	3-9%	6-12%	Better paywall, stronger local accuracy
B2C ARPU	EUR 49-79	EUR 70-100	EUR 100-140	Pro+ and annual plan mix
Churn annual	35%	25%	15%	Season pass, club bundles, retention alerts
Club/school sales	Slow	Moderate	Fast	Founder-led sales and ambassadors
Data cost/user	High	Controlled	Optimized	Caching and forecast cube tiling

12. Revenue Forecast

Year	B2C	Club	School	Event	API	Total revenue
Y1	EUR 10,500	EUR 2,520	EUR 2,160	EUR 0	EUR 0	EUR 15,180
Y2	EUR 102,500	EUR 21,000	EUR 21,600	EUR 6,000	EUR 6,000	EUR 157,100
Y3	EUR 483,000	EUR 72,000	EUR 84,000	EUR 24,000	EUR 28,800	EUR 691,800
Y5	EUR 2,250,000	EUR 255,000	EUR 345,600	EUR 120,000	EUR 144,000	EUR 3,114,600

The base case is not a promise. It is a planning model. The first real commercial checkpoint is whether the product can produce weekly active use and a 3-5% paid conversion in one or two launch-dense regions.

Scenario model

Scenario	Year 3 revenue	Assumptions	Investor interpretation
Low	EUR 200k-350k	Slow conversion, mostly B2C, limited B2B	Useful lifestyle/bootstrapped product but not venture-scale yet
Base	EUR 600k-750k	Healthy B2C plus emerging club/school traction	Potential seed-stage SaaS with strong niche
High	EUR 1.2M-2.0M	Strong regional adoption, B2B, events and station network	Attractive vertical SaaS / data network opportunity

Revenue quality

B2C subscriptions validate user demand. Club, school and event revenue improves revenue quality because it is less sensitive to individual churn and creates local distribution. API and rescue revenue should be added only after data quality and reliability are strong enough to support partner SLAs.

13. Cost Model and Gross Margin

The platform has three major cost classes: product/team cost, weather/data/infrastructure cost, and go-to-market/support cost. Variable infrastructure costs can be controlled through forecast-cube caching, map-tile precomputation, regional rollout and careful API-rate design.

Cost category	MVP monthly estimate	Year 2 monthly estimate	Notes
Cloud compute and storage	EUR 300-1,000	EUR 2,000-8,000	Weather processing, API, DB, maps, tracking
Weather/API/data licenses	EUR 0-1,500	EUR 1,000-10,000	Depends on commercial data rights and premium sources
Monitoring/logging/CDN	EUR 100-500	EUR 500-2,500	Scale with traffic and map usage
App store/payment fees	15-30% of in-app revenue	15-30% of in-app revenue	May vary by region and provider
Support/community moderation	Founder time	EUR 2,000-8,000	Required for local trust and B2B accounts
Team payroll	Founder/bootstrap	Largest expense	Depends on hiring model

Gross margin target

Target software gross margin should exceed 75% after the early phase. Weather-station hardware can have lower margin but improves the data moat. Avoid scaling expensive raw model processing globally before conversion is proven region by region.

Cost-control architecture

- Process forecast data by region and priority sites first, not globally on day one.
- Cache site forecasts and tiles aggressively; do not recompute per user request.
- Separate free-tier limits from paid-tier high-resolution data.
- Use open/public data where license permits, but budget for commercial data if needed.
- Monitor API cost per active user and cost per paid user as first-class KPIs.

14. Unit Economics

B2C unit economics

Metric	Target	Rationale
Annual ARPU	EUR 70-100	Blend of Pro and Pro+
Annual churn	<25% after product-market fit	Seasonal app but high annual utility
Gross margin	>75%	Required for scalable SaaS
CAC payback	<6 months for paid ads; near-zero for organic/club channels	Niche market cannot support expensive broad ads
LTV:CAC	>3:1	Minimum efficient paid growth target

B2B unit economics

Segment	Annual contract value target	Sales cost model	Expansion potential
Club	EUR 500-1,200/year	Founder-led then self-serve	Add station dashboard and member packs
School	EUR 1,500-3,500/year	Direct sales	Add scheduling, student analytics, tandem ops
Event	EUR 500-3,000/event	Seasonal direct sales	Add tracking, task planning, safety board
API	EUR 1,200-12,000/year	Partner sales	Usage-based expansion

Retention mechanics

- Seasonal annual renewal reminders tied to flying season start.
- Personal flight history and debriefs create stored value.
- Favorite-site alert rules create habit and switching cost.
- Club/school accounts create group dependency and internal workflows.
- Local digital twins improve with use, creating accuracy-based retention.

15. Funding and Milestone Strategy

Funding strategy should follow evidence, not document volume. The app should first prove that pilots trust and repeatedly use launch-specific recommendations. Then it should prove paid conversion. Then it should prove B2B adoption and regional expansion.

Stage	Funding type	Milestone	Evidence required
Prototype	Self-funded / grants	Clickable app + 10 launch playbooks	Pilot interviews and UI validation
MVP	Angel/pre-seed or bootstrapped	Live forecasts, flyability, alerts for 1-2 regions	Weekly active use and forecast-vs-reality logs
Commercial beta	Pre-seed/seed	Paid Pro/Pro+ and first clubs/schools	MRR, retention, conversion, testimonials
Regional scale	Seed	Multi-region expansion and B2B pipeline	CAC/LTV, gross margin, B2B sales velocity

Milestone gates

Gate	Pass criteria	Why it matters
Trust gate	Pilots agree score explanations match reality in >80% reviewed cases	Trust is the product
False-green gate	No unresolved critical false-green incidents in beta safety review	Safety/liability protection
Conversion gate	3-5% free-to-paid in target region	Proves monetization
Retention gate	>35% monthly active retention in flying season	Proves habit
B2B gate	5+ paying org accounts	Proves expansion beyond individual pilots

16. Operating Plan and Team

Team phases

Phase	Team	Primary work
Prototype	Founder + contractor	Mobile/web prototype, site data, weather API integration
MVP	2-4 people	Backend, mobile, weather engine, UX, QA, beta ops
Commercial beta	5-8 people	Product, backend, mobile, DevOps, community, B2B sales
Scale	10-20 people	ML/weather, SRE, support, partnerships, sales, compliance

Critical roles

Role	Why critical	When needed
Weather/GIS engineer	Forecast cubes, terrain, model interpolation	MVP or early V1
Mobile engineer	Map, offline, tracking, alert UX	MVP
Backend/platform engineer	APIs, DB, queues, billing, entitlements	MVP
Paragliding domain lead	Rules, launch playbooks, safety review	Immediately
Community manager	Clubs, reports, ambassadors	Commercial beta
B2B sales founder/operator	Clubs, schools, events	Commercial beta
Security/privacy owner	Location, tracking, GDPR, billing	Before launch

Operating cadence

- Weekly forecast validation review during flying season.
- Monthly safety-review meeting for false-green and false-red cases.
- Monthly business dashboard review: MRR, conversion, churn, CAC, active sites.
- Quarterly roadmap and pricing review.

17. KPI and Analytics System

The business needs a combined product, safety and financial KPI system. A weather-decision app can show high usage but poor trust, or high trust but poor monetization. Both must be measured.

KPI category	Metric	Target direction
Acquisition	New registered users by region	Increase
Activation	User adds first favorite site	>60% of registrations
Habit	Weekly active pilots in flying season	Increase
Trust	Forecast-vs-reality agreement score	Improve
Safety	Critical false-green incidents	Zero tolerance / formal review
Monetization	Free-to-paid conversion	3-9% by maturity
Revenue	MRR and ARR	Increase
Retention	Annual renewal rate	>70-80% mature
B2B	Club/school pipeline and close rate	Increase
Infrastructure	Cost per active user	Decrease

Analytics event taxonomy

site_viewed

favorite_site_added

flyability_timeline_opened

model_comparison_opened

ai_briefing_requested

alert_rule_created

drive_recommendation_viewed

paywall_viewed

subscription_started

report_submitted

forecast_reality_compared

tracking_session_started

no_go_blocker_displayed

Decision-quality metrics

The most important non-financial metric is decision quality: when the app said GO/MAYBE/NO-GO, did actual conditions and expert review support that decision? This should be measured per site, forecast model, hour, season and pilot level.

18. Sales Motions and Playbooks

B2C self-serve sales

Moment	Message	Offer
User checks 3+ sites in one week	You are actively planning flights. Unlock unlimited favorites and advanced alerts.	Pro annual trial
User opens model comparison locked screen	Forecast disagreement matters today. Unlock model confidence.	Pro+
User requests AI briefing repeatedly	Save time with daily briefings and XC planning.	Pro+
User creates alert limit reached	Never miss a good window at your favorite sites.	Pro

Club sales playbook

32. Identify clubs managing sites with active weather discussions.
33. Offer a 30-day club dashboard with 3-5 priority sites.
34. Add one trusted club weather officer as admin.
35. Enable verified reports and daily briefing links.
36. Collect testimonial after three usable flying weekends.
37. Convert to annual club plan with member discount codes.

School sales playbook

38. Target schools/tandem operators with repeated weather-dependent operations.
39. Offer instructor dashboard and daily PDF/WhatsApp briefing export.
40. Demonstrate student-level suitability and hard-blocker explanations.
41. Provide season package with support and custom launch limits.
42. Upsell weather station integration and branded student onboarding.

19. Product Roadmap to Revenue

Release	Commercial objective	Features	Revenue target
MVP	Trust and habit	Launch pages, flyability, alerts, basic AI briefing, reports	Pilot beta, no or low revenue
V1	B2C monetization	Pro/Pro+, model comparison, drive engine, XC preview	First EUR 5k-20k MRR
V1.5	Club/school revenue	Dashboards, verified reports, export briefings	First 20 org accounts
V2	Defensibility	Digital twins, forecast correction, weather stations	Higher retention and Pro+ conversion
V3	Operations platform	Events, rescue, API, marketplace	Diversified revenue

Revenue-first backlog priorities

- Drive-or-dont-drive recommendation because it connects directly to saved travel cost.
- Advanced alerts because they create habit and justify Pro.
- Model confidence and disagreement because pilots already compare models manually.
- AI briefing because it saves cognitive work but must be strictly grounded.
- Club dashboard because clubs are distribution nodes and B2B revenue anchors.

20. Legal, Data Licensing and Safety Liability

This business plan requires a dedicated legal/data-licensing workstream before commercial launch. Weather, map, DEM, airspace, app-store and payment terms can materially affect what can be monetized, cached, redistributed or exposed through an API.

Area	Commercial risk	Required action
Weather data	Some sources may restrict commercial redistribution or caching	Review source licenses and attribution obligations
Map tiles	Tile providers may restrict heavy commercial use	Use paid/compliant map provider or self-host allowed tiles
Airspace data	Safety-critical and licensing-sensitive	Use licensed/attribution provider and update logs
Location tracking	GDPR/privacy sensitivity	Explicit consent, retention controls, private-by-default tracking
AI recommendations	Hallucination and liability risk	Grounded context only, no invented weather, mandatory disclaimers
App stores	Subscription rules and commissions	Design billing architecture for Apple/Google/Stripe
Rescue mode	Sensitive incident data	Strict access control, audit logs, legal agreements

Safety marketing rule

Do not market the product as guaranteeing safety. The product provides weather analysis, risk flags and decision support. All recommendations must show confidence, model disagreement and hard safety blockers. If data is missing, stale or contradictory, the app should say NO DATA or MAYBE, not force a green result.

Commercial copy guardrails

Allowed	Avoid
"Decision support for paragliding weather"	"Guarantees safe flying"
"Forecast confidence and risk briefing"	"This launch is safe" without conditions
"GO/MAYBE/NO-GO based on available data"	"AI decides if you can fly"
"Pilot remains responsible for final decision"	Any wording that shifts legal responsibility to the app

21. Risk Register

Risk	Likelihood	Impact	Mitigation
False-green recommendation damages trust	Medium	Very high	Safety blockers, QA, conservative scoring, expert review
Data license prevents commercial use	Medium	High	License review before scaling; fallback sources
Weather accuracy not better than incumbents	Medium	High	Focus on decision UX and local correction, not raw forecast superiority
Low paid conversion	Medium	High	Pricing experiments, stronger premium triggers, club bundles
B2B sales slower than expected	Medium	Medium	Start founder-led, offer trials, use ambassadors
Infrastructure costs grow too fast	Medium	Medium	Regional processing, caching, quotas, free-tier limits
Community reports become noisy	High	Medium	Reputation, verification, moderation, anomaly detection
Privacy issue with live tracking	Medium	High	Private by default, consent, audit, retention controls
AI hallucination	Medium	High	Grounding, refusal rules, output schemas, test suite

Top three existential risks

- 43. The app gives a wrong green recommendation and loses community trust.
- 44. The product remains a nice weather viewer but does not produce paid conversion.
- 45. Data licensing or infrastructure cost makes the business model uneconomic.

22. Investor Narrative

Investor narrative should focus on vertical decision intelligence, safety-critical habit, proprietary local weather correction and B2B expansion. Avoid presenting the company as a generic weather app because that invites direct comparison with well-funded incumbents and lowers perceived defensibility.

Pitch structure

- 46. Problem: pilots make high-consequence weather decisions using fragmented tools.
- 47. Solution: paragliding-specific decision intelligence with GO/MAYBE/NO-GO, confidence and local launch models.
- 48. Why now: mobile maps, open forecast APIs, sensor data, AI summarization and community data make vertical weather intelligence possible.
- 49. Moat: launch digital twins, forecast error history, verified reports, clubs/schools and weather stations.
- 50. Business model: B2C subscriptions plus B2B clubs, schools, events, API and hardware.
- 51. Milestones: trust, retention, paid conversion, B2B accounts, regional expansion.

What investors will challenge

Investor question	Required answer
How big is the market?	Use bottom-up active-pilot and B2B segment evidence; do not overclaim.
Why will pilots pay?	Show saved trips, better planning, community trust and Pro conversion data.
Why wont Windy copy it?	Windy can copy layers, but not local digital twins, community trust and B2B workflows quickly.
What about liability?	Conservative decision support, QA, disclaimers, no-data states and safety review process.
What is the moat?	Historical local corrections, reports, station network and launch playbooks.

23. Board Dashboard Template

Dashboard section	Metric	Current	Target	Owner
Growth	Registered users			Growth
Activation	% adding first favorite site		>60%	Product
Engagement	Weekly active users			Product
Trust	Forecast-vs-reality score		Improving	Weather
Safety	Critical false-green cases		0 unresolved	Safety
Revenue	MRR / ARR			CEO
Conversion	Free-to-paid %		3-9%	Product/Growth
Retention	Annual renewal %		>70%	Product
B2B	Club/school accounts			Sales
Cost	Infra cost per active user		Downward	DevOps

Monthly decision questions

- Which sites had the biggest forecast error and why?
- Which premium feature drove the most conversions?
- Which acquisition channel produced retained active pilots?
- Did any safety blocker fail or create false confidence?
- Which region is ready for expansion?

24. MVP Commercial Launch Plan

90-day commercial MVP plan

Week range	Goal	Deliverables
1-2	Define beachhead and launch playbooks	10 sites, wind limits, local experts, source list
3-4	Implement public site pages and basic forecasts	SEO pages, site details, basic map
5-6	Add flyability and alerts	Score API, timeline, push/email alerts
7-8	Add reports and beta community	Pilot reports, verified users, trust feedback
9-10	Add Pro paywall and entitlements	Stripe/app-store product IDs, feature locks
11-12	Run paid beta and review	Conversion test, QA review, pricing feedback

MVP commercial acceptance criteria

Area	Acceptance criterion
Trust	At least 20 expert-reviewed days across 10 sites with documented forecast-vs-reality results
Engagement	At least 100 weekly active beta users in flying season
Reports	At least 50 verified pilot observations
Monetization	At least 20 paying users or explicit commitments
Safety	No unresolved critical false-green cases
B2B	At least 2 club/school pilots using dashboard manually or in prototype

25. Appendices and Source Notes

Source notes used for business assumptions

Source	URL	Use in plan
Windy subscription/features	https://www.windy.com/subscription	Premium feature context such as forecast steps, forecast horizon and radar/satellite loops.
Ventusky product/premium pages	https://www.ventusky.com/ and https://my.ventusky.com/premium/features	General weather layer/model positioning and premium long-range/model comparison context.
XCTrack official site	https://xctrack.org/ and https://www.xcontest.org/xctrack/	Flight-computer, XC, competition, airspace and tracking feature benchmark.
Paragliding Map	https://www.paraglidingmap.com/	Site database, stations and webcams benchmark.
FAI/CIVL	https://www.fai.org/commission/civil	Competition ecosystem and official free-flight activity context.
Paragliding equipment market reports	DataIntelto, Deep Market Insights, Research and Markets snippets	Directional market-growth context only; not treated as a precise TAM.
Stripe Billing	https://docs.stripe.com/billing/subscriptions/overview	Subscription lifecycle and billing architecture context.
Apple in-app subscriptions	https://developer.apple.com/app-store/subscriptions/	iOS subscription management and server notification context.
Google Play Billing subscriptions	https://developer.android.com/google/play/billing/subscriptions	Android subscription lifecycle context.
GDPR legal text / EDPB guidance	https://gdpr-info.eu/ and https://www.edpb.europa.eu/	Privacy and location-data governance context.

Assumption register

Assumption	Validation method
Pilots will pay for decision confidence, not just more weather layers	Landing page A/B test and in-app paywall conversion
Clubs and schools are viable B2B channels	Founder-led sales calls and pilot dashboards
Local correction improves trust over time	Forecast-vs-reality logs and model skill tracking
Annual plans fit seasonal sport behavior	Subscription plan experiment monthly vs annual
Safety trust is more valuable than aggressive green recommendations	Pilot interviews and incident review board

Final recommendation

The next business action is not to raise money from a large document. The next action is to validate trust and conversion in a narrow region: build the MVP with 10 high-quality launch playbooks, run forecast-vs-reality validation for a season window, recruit local pilots and clubs, and test Pro/Pro+ conversion. If the app proves that it saves trips and improves decision confidence, the rest of the platform can scale into B2B, events, API and digital twins.